Particle-in-a-Box Residuals and Identifiability

Provisional Experiment Report and Draft Appendix D

`ksdft2effmass` research workspace

Evidence status. Calculated illustrative numerical experiment and provisional manuscript draft. This document is not a reviewed manuscript, released research result, semiconductor calculation, scientific validation result, or uncertainty-quantification study.

Abstract

The one-dimensional particle in a box provides a closed-form setting in which a continuum operator, its domain, a finite representation, spectral retention, and reduced-model interpretation can be kept distinct. This report tests three matrix residuals that could otherwise be conflated under the label of an effective potential. Consistent compression of the Dirichlet Hamiltonian and kinetic operator gives zero potential. Compressing only the Hamiltonian produces the negative discarded kinetic sector. Subtracting a cyclic finite-space reference produces a boundary-realization difference.

Grid-refinement calculations over $N = 8, 16, 32, 64, 128, 256$ interior points verify second-order convergence for fixed eigenvalue indices. A full-spectrum sweep shows that this convergence is not uniform when mode index grows with grid dimension. Frobenius, spectral, and maximum-entry norm studies show that a residual can become small in an aggregate same-grid norm while remaining significant in a worst-case norm. Finally, two distinct exact decompositions of the same retained Hamiltonian and four model-class fits demonstrate the central identifiability result: a reduced Hamiltonian does not select a unique kinetic–potential decomposition. A reference operator and admissible model class must be specified before a residual receives a physical interpretation.

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1 Purpose

The purpose is not to solve a difficult spectrum. The spectrum is known in closed form. The purpose is to separate five ingredients that are easily mixed in a more complicated reduction:

1. the continuum Hamiltonian and its domain;
2. the finite numerical representation;
3. the full and retained state spaces;
4. the projection and embedding maps; and
5. the physical interpretation assigned to a residual.

The controlled example asks what can be concluded from a subtraction such as

$$H_{\mathrm{eff}} - T. \tag{1}$$

The answer depends on which Hamiltonian and kinetic reference are used, whether they act on the same space, whether both passed through the same reduction map, and which potential model class is admissible. Numerical convergence alone does not answer those questions.

2 Continuum operator and finite realization

For a particle of mass m on $(0, L)$, the continuum reference is the Dirichlet realization

$$\hat{H}_{\mathrm{box}} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}, \qquad \mathcal{D}(\hat{H}_{\mathrm{box}}) = H^2(0, L) \cap H_0^1(0, L). \tag{2}$$

The boundary conditions are $\psi(0) = \psi(L) = 0$. The eigenpairs are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\Big(\frac{n\pi x}{L}\Big), \qquad E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}. \tag{3}$$

The interior physical potential is zero. Confinement is represented by the domain of the Dirichlet kinetic operator rather than by a finite-valued multiplication potential inside the interval. This distinction is essential: the differential expression alone does not define the self-adjoint operator, and changing the boundary condition changes the operator even when the formal kinetic expression is unchanged [1, 2].

For N interior points and spacing $h = L/(N + 1)$, the finite coordinate space is $\mathcal{V}_h \cong \mathbb{R}^N$. The centered second-order realization is

$$\mathbf{H}_h = \frac{\hbar^2}{2mh^2} \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}. \tag{4}$$

The calculated experiments use the dimensionless convention $L = m = \hbar = 1$. The centered stencil is a second-order finite-difference approximation in the usual local truncation sense [3]. For this particular Toeplitz matrix, the complete discrete spectrum is also available analytically:

$$E_{n,h} = \frac{2\hbar^2}{mh^2} \sin^2\left(\frac{n\pi}{2(N+1)}\right), \qquad n = 1, \ldots, N. \tag{5}$$

This formula supplies an oracle independent of the numerical eigensolver and makes it possible to distinguish discretization error from algebraic solution error throughout the report.

3 Spectral retention and state-space maps

Let $\mathbf{\Phi}_M \in \mathbb{R}^{N\times M}$ contain the lowest M orthonormal eigenvectors of $\mathbf{H}_h$. Define

$$\mathbf{P}_M = \mathbf{\Phi}_M \mathbf{\Phi}_M^T, \qquad \mathbf{Q}_M = \mathbf{I} - \mathbf{P}_M. \tag{6}$$

Two representations of the retained operator are needed:

$$\mathbf{H}_h^{(P_M)} = \mathbf{P}_M \mathbf{H}_h \mathbf{P}_M \in \mathbb{R}^{N\times N}, \tag{7}$$

$$\mathbf{H}_{\text{red}} = \mathbf{\Phi}_M^T \mathbf{H}_h \mathbf{\Phi}_M \in \mathbb{R}^{M\times M}. \tag{8}$$

The first is an embedded rank-M matrix on the full numerical space. The second is the same retained operator in retained spectral coordinates. Their dimensions differ, so an embedding must be declared before comparison.

For the retained $N = 8$, $M = 3$ calculation, the numerical checks were:

Table 1: Finite-representation diagnostics.

Diagnostic	Calculated value
Discrete spectrum maximum absolute oracle error	8.53×10^{-14}
Spectral reconstruction Frobenius error	3.04×10^{-13}
Projector idempotency Frobenius error	8.48×10^{-16}
Retained-coordinate Frobenius error	2.46×10^{-14}
Retained embedding Frobenius error	1.02×10^{-14}

4 Three inequivalent residuals

4.1 Consistently compressed physical potential

In this model, $\mathbf{H}_h = \mathbf{T}_{\text{D},h}$. Applying the same compression to both terms gives

$$\mathbf{P}_M \mathbf{H}_h \mathbf{P}_M - \mathbf{P}_M \mathbf{T}_{\text{D},h} \mathbf{P}_M = \mathbf{0}. \tag{9}$$

The calculated Frobenius norm is exactly zero. This is the consistently reduced physical potential for the declared box model.

4.2 Compressed Hamiltonian minus uncompressed kinetic operator

If only the Hamiltonian is compressed,

$$\widetilde{\mathbf{R}}_h = \mathbf{P}_M \mathbf{H}_h \mathbf{P}_M - \mathbf{T}_{\mathrm{D},h}, \tag{10}$$

then spectral commutation gives

$$\widetilde{\mathbf{R}}_h = -\mathbf{Q}_M \mathbf{T}_{\mathrm{D},h} \mathbf{Q}_M. \tag{11}$$

For $N = 8$ and $M = 3$, the calculated Frobenius norm is 270.977, and the Frobenius error against the negative discarded-sector oracle is 8.75×10^{-14}. The nonzero matrix is caused by unmatched reduction. It is not the physical box potential.

4.3 Difference between boundary realizations

A cyclic closure was declared as a comparison reference on the same finite coordinate space. The difference

$$\mathbf{B}_h = \mathbf{H}_{\mathrm{D},h} - \mathbf{H}_{\mathrm{cyclic},h} \tag{12}$$

contains two corner couplings. At $N = 8$, its Frobenius norm is 57.276. This is a finite boundary-realization difference relative to the declared cyclic reference, not a representation-independent continuum potential.

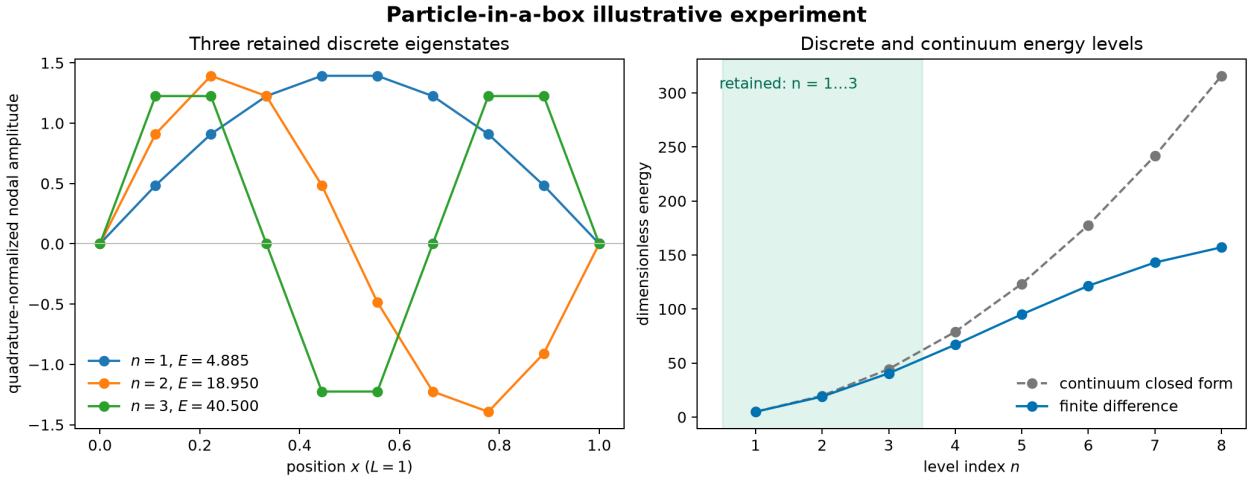


Figure 1: The three retained states and the finite and continuum spectra. The finite grid underestimates increasingly high continuum energies.

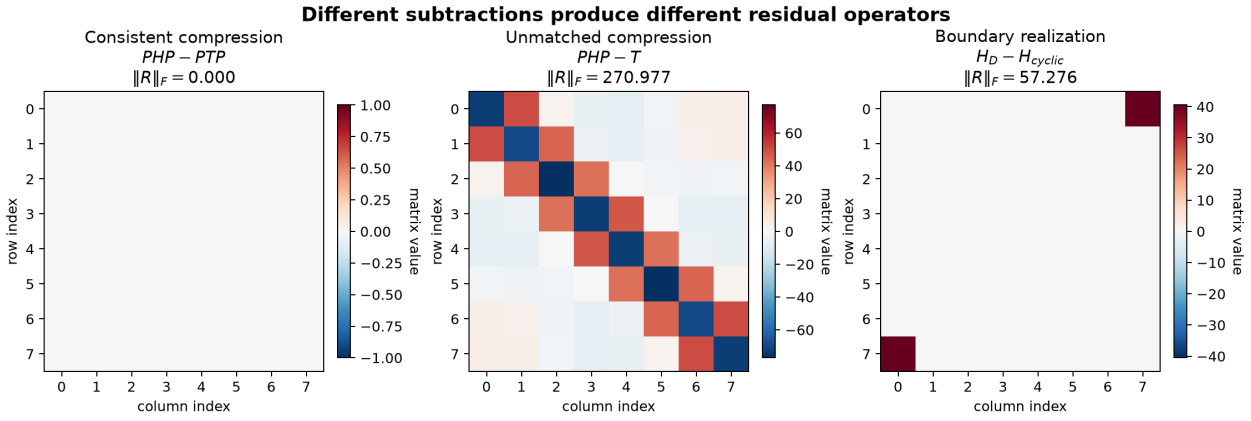


Figure 2: The three residual constructions. Identical subtraction syntax does not imply identical mathematical or physical meaning.

5 Fixed-mode convergence

The convergence study is retained because the finite representation must be verified independently of residual interpretation. The grid sequence is $N = 8, 16, 32, 64, 128, 256$. For each fixed mode, the relative energy error is

$$\varepsilon_n(N) = \frac{|E_{n,h} - E_n|}{E_n}. \tag{13}$$

Writing

$$z_{n,N} = \frac{n\pi}{2(N+1)} = \frac{n\pi h}{2L}, \tag{14}$$

the exact ratio between the discrete and continuum eigenvalues is

$$\frac{E_{n,h}}{E_n} = \left(\frac{\sin z_{n,N}}{z_{n,N}}\right)^2. \tag{15}$$

For fixed n and $h \to 0$, Taylor expansion gives

$$\varepsilon_n(N) = \frac{z_{n,N}^2}{3} + \mathcal{O}(z_{n,N}^4) = \frac{n^2\pi^2}{12L^2}h^2 + \mathcal{O}(h^4). \tag{16}$$

Thus the observed order-two trend is not merely empirical; it is the leading term predicted by the exact dispersion relation. The coefficient grows as n^2, explaining why higher fixed modes require finer grids before entering the same asymptotic regime.

5.1 Low fixed modes

Table 2: Relative energy errors for the first three fixed modes.

N	$n = 1$	$n = 2$	$n = 3$
8	1.0113×10^{-2}	3.9962×10^{-2}	8.8109×10^{-2}
16	2.8427×10^{-3}	1.1332×10^{-2}	2.5352×10^{-2}
32	7.5502×10^{-4}	3.0174×10^{-3}	6.7788×10^{-3}
64	1.9465×10^{-4}	7.7842×10^{-4}	1.7508×10^{-3}
128	4.9423×10^{-5}	1.9768×10^{-4}	4.4474×10^{-4}
256	1.2452×10^{-5}	4.9809×10^{-5}	1.1207×10^{-4}

The final observed orders are 1.99998, 1.99991, and 1.99981 for $n = 1, 2, 3$, respectively. This is the expected second-order behavior for fixed resolved modes.

5.2 Fixed higher modes

A fixed higher mode enters when it exists on the grid. The series starts at $N = 8$ for $n = 8$, at $N = 16$ for $n = 16$, and at $N = 32$ for $n = 32$.

The initial points are under-resolved and pre-asymptotic. Each fixed mode moves toward second-order behavior as the number of grid points per wavelength increases.

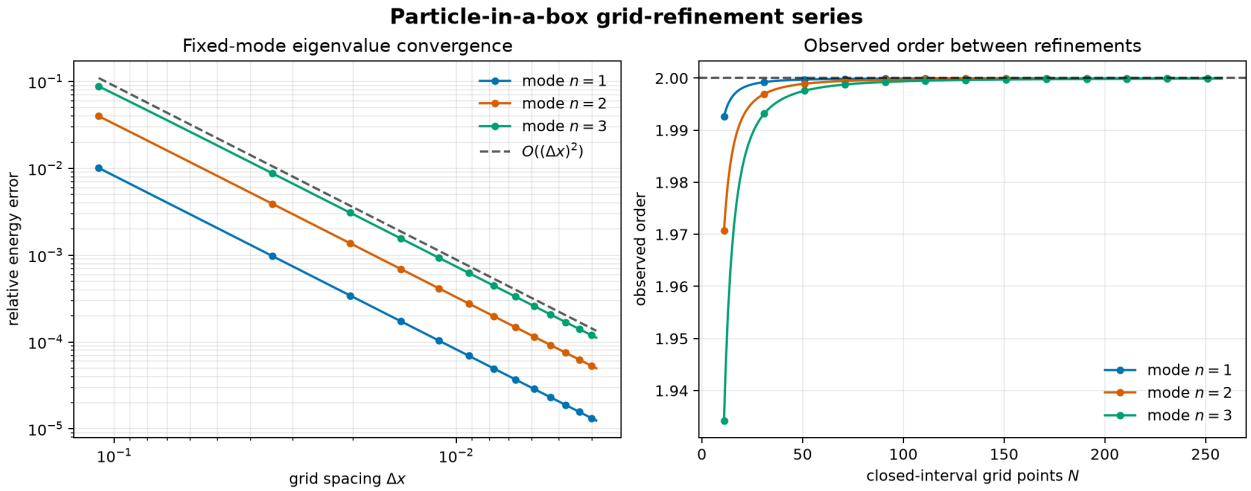


Figure 3: Fixed low-mode relative errors and observed refinement orders.

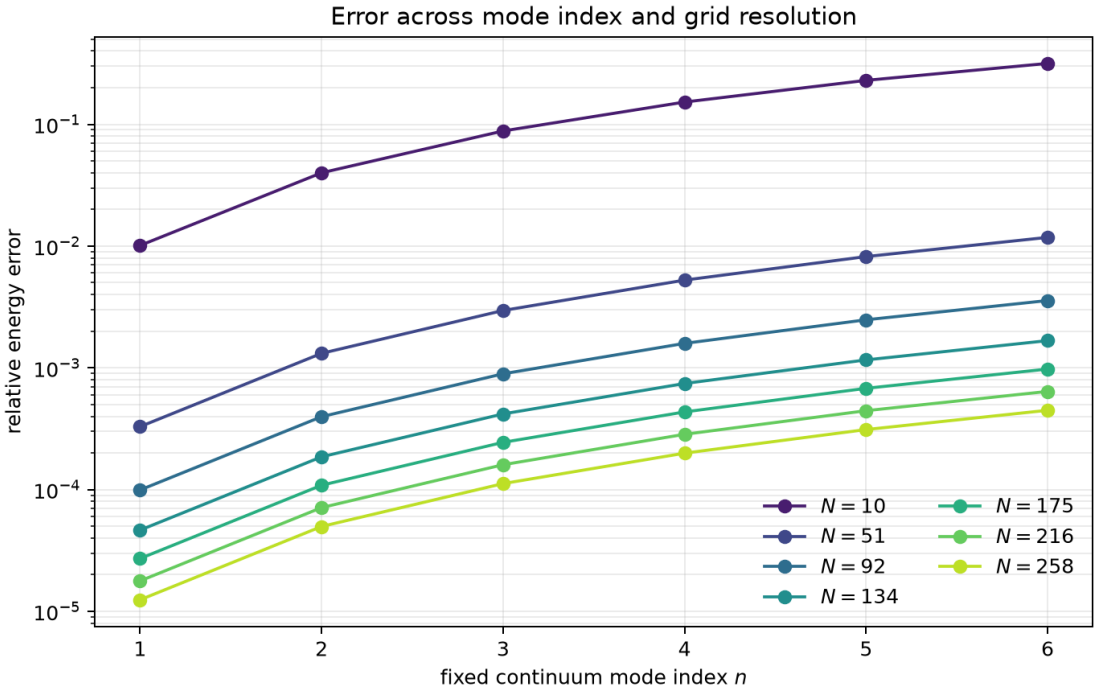


Figure 4: Relative energy error across the first six fixed mode indices and the full grid-refinement series.

Table 3: Convergence of fixed higher eigenvalues.

Mode	First-grid relative error	$N = 256$ relative error	Final order
8	5.0253×10^{-1}	7.9670×10^{-4}	1.99863
16	5.4637×10^{-1}	3.1837×10^{-3}	1.99451
32	5.6997×10^{-1}	1.2686×10^{-2}	1.97805

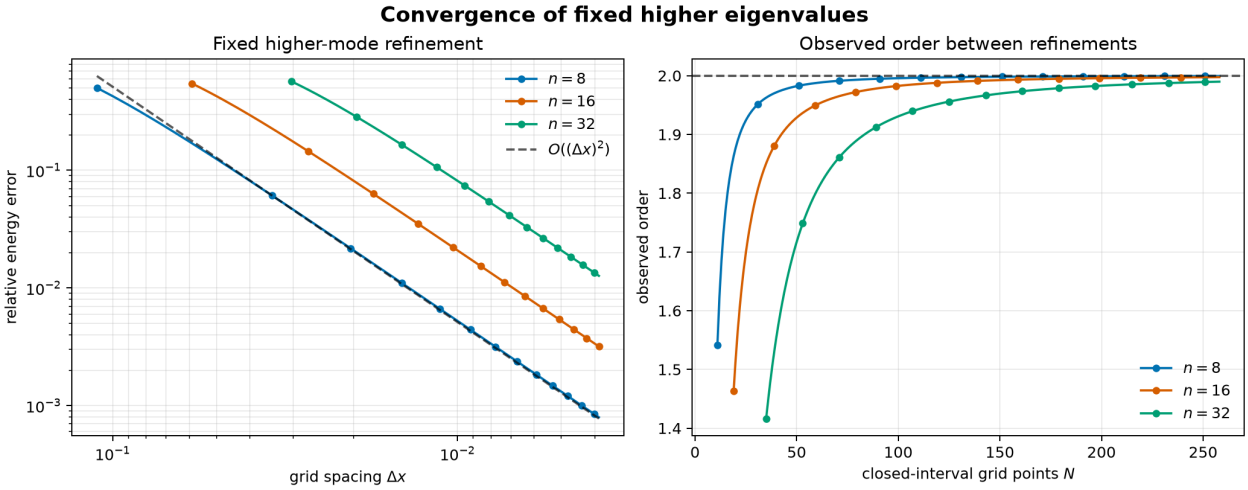


Figure 5: Fixed higher-mode convergence and approach to second order.

6 Full-spectrum eigenpair sweep

Fixed-index convergence is not uniform spectral convergence. To expose that distinction, the full discrete spectra were calculated for every grid from $N = 8$ through $N = 256$. When the fractional mode index

$$q = \frac{n}{N+1} \tag{17}$$

is held approximately fixed, relative eigenvalue error remains substantial as N grows. Indeed, the exact eigenvalue ratio becomes

$$\frac{E_{n,h}}{E_n} = \left[\frac{\sin(\pi q/2)}{\pi q/2}\right]^2. \tag{18}$$

At fixed nonzero q, this expression is independent of h. Consequently the relative error has a nonzero refinement limit. The full-spectrum curve collapse is therefore an analytical feature of the stencil rather than a failure of the eigensolver. Fixed-index convergence and convergence at fixed fractional mode index are different limiting statements and must not be interchanged.

Table 4: Representative full-spectrum errors at $N = 256$.

q	Mode	Relative energy error
0.249	64	4.998×10^{-2}
0.498	128	1.881×10^{-1}
0.751	193	3.858×10^{-1}
0.996	256	5.916×10^{-1}

The near-collapse of the curves as functions of q is a finite-difference dispersion result. Refining the grid improves each fixed physical mode, but it does not make the entire growing discrete spectrum uniformly accurate.

At $N = 256$, the maximum nodal overlap defect against sampled continuum sine vectors is 4.00×10^{-15}, and the maximum scaled algebraic eigenpair residual is 7.14×10^{-16}. These values show that the discrete eigenpairs were solved accurately and agree with sine vectors at the nodes. They do not measure error of an interpolated continuum wavefunction.

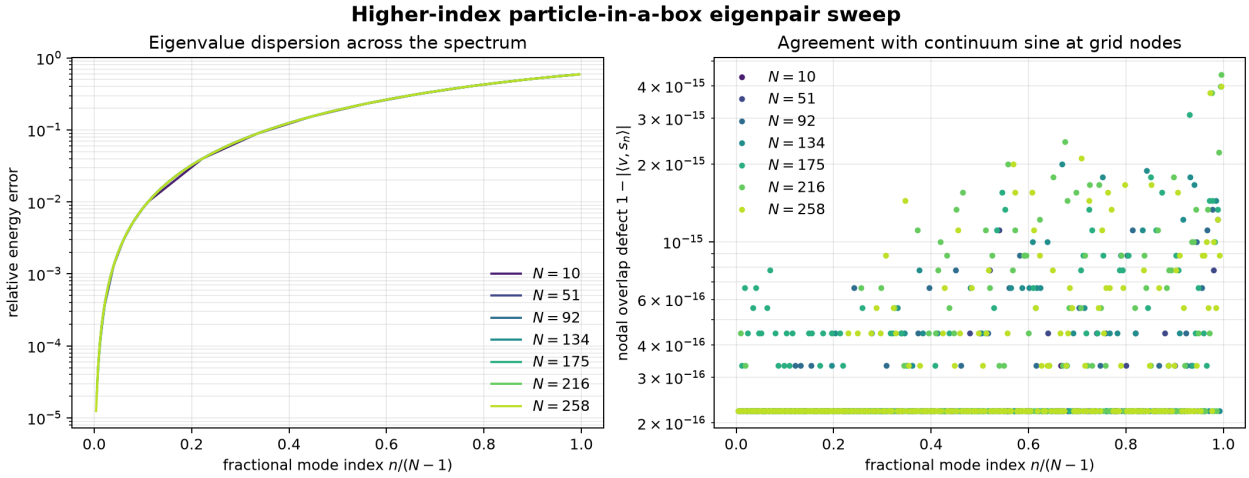


Figure 6: Full-spectrum eigenvalue dispersion and nodal overlap diagnostics.

7 Dependence on matrix norm

A residual does not have one universal magnitude. For a residual $\mathbf{R}_h$, the same-grid ratios are

$$\frac{\|\mathbf{R}_h\|_F}{\|\mathbf{H}_h\|_F}, \qquad \frac{\|\mathbf{R}_h\|_2}{\|\mathbf{H}_h\|_2}, \qquad \frac{\|\mathbf{R}_h\|_{\max}}{\|\mathbf{H}_h\|_{\max}}. \tag{19}$$

Raw norms are retained but are not interpreted as convergence metrics across changing spaces because matrix dimension and the h^{-2} kinetic scale both change.

The norm trends can be derived explicitly. Let

$$c_h = \frac{\hbar^2}{2mh^2}. \tag{20}$$

For the Dirichlet matrix,

$$\|\mathbf{H}_h\|_F = c_h\sqrt{6N-2}, \qquad \|\mathbf{H}_h\|_{\max} = 2c_h, \tag{21}$$

and

$$\|\mathbf{H}_h\|_2 = 4c_h \cos^2\left(\frac{\pi}{2(N+1)}\right). \tag{22}$$

The boundary residual consists only of the two corner entries of magnitude c_h. Hence

$$\|\mathbf{B}_h\|_F = \sqrt{2}\, c_h, \qquad \|\mathbf{B}_h\|_2 = c_h, \qquad \|\mathbf{B}_h\|_{\max} = c_h. \tag{23}$$

Its normalized ratios therefore satisfy

$$\frac{\|\mathbf{B}_h\|_F}{\|\mathbf{H}_h\|_F} = \frac{1}{\sqrt{3N-1}} \longrightarrow 0, \tag{24}$$

$$\frac{\|\mathbf{B}_h\|_2}{\|\mathbf{H}_h\|_2} = \frac{1}{4\cos^2[\pi/(2(N+1))]} \longrightarrow \frac{1}{4}, \qquad \frac{\|\mathbf{B}_h\|_{\max}}{\|\mathbf{H}_h\|_{\max}} = \frac{1}{2}. \tag{25}$$

The apparently different graphical trends are thus exact consequences of the different geometries encoded by the norms: aggregate Hilbert–Schmidt size, worst-case Euclidean action, and largest coordinate entry.

7.1 Unmatched compression

Table 5: Same-grid norm ratios for unmatched compression.

N	Frobenius	Spectral	Maximum entry
8	0.9865	1.0000	0.9510
16	0.9994	1.0000	0.9940
32	1.0000	1.0000	0.9998
64	1.0000	1.0000	1.0000
128	1.0000	1.0000	1.0000
256	1.0000	1.0000	1.0000

A fixed three-state retention discards nearly all of the increasingly resolved operator norm. In the discrete eigenbasis, the unmatched residual has three zero eigenvalues and the values $-E_{4,h}, \ldots, -E_{N,h}$. It follows directly that its spectral-norm ratio is one. Its squared Frobenius ratio is

$$\frac{\|\widetilde{\mathbf{R}}_h\|_F^2}{\|\mathbf{H}_h\|_F^2} = 1 - \frac{\sum_{n=1}^{M} E_{n,h}^2}{\sum_{n=1}^{N} E_{n,h}^2}, \tag{26}$$

which approaches one for fixed M as N grows. The trend therefore records what was excluded by the fixed-rank retention; it is not evidence for an emergent potential and does not approach zero.

7.2 Boundary realization

Table 6: Same-grid norm ratios for the boundary realization.

N	Frobenius	Spectral	Maximum entry
8	0.2085	0.2578	0.5000
16	0.1459	0.2521	0.5000
32	0.1026	0.2506	0.5000
64	0.0724	0.2501	0.5000
128	0.0511	0.2500	0.5000
256	0.0361	0.2500	0.5000

The Frobenius ratio decreases because two boundary couplings become small relative to the aggregate norm of an $N \times N$ Hamiltonian. The spectral ratio approaches 0.25, so worst-case Euclidean action remains a fixed fraction of the Hamiltonian scale. The maximum-entry ratio remains 0.5 in the declared grid basis. These statements are compatible because the norms answer different questions.

The full-eigensystem algebraic residual matrix

$$\mathbf{H}_h\mathbf{V} - \mathbf{V}\mathbf{\Lambda} \tag{27}$$

has same-grid norm ratios below 10^{-15} throughout the series. This is an eigensolver diagnostic only.

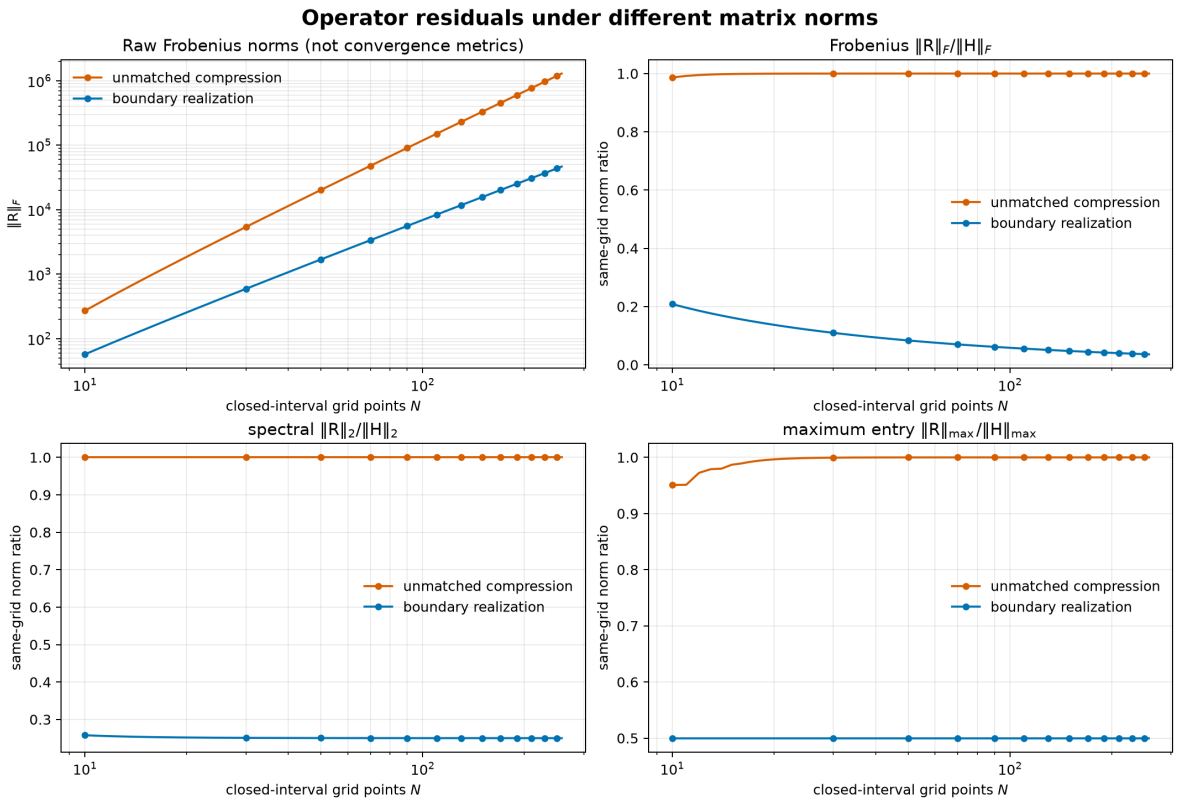


Figure 7: Raw Frobenius norms and same-grid-normalized Frobenius, spectral, and maximum-entry residual norms.

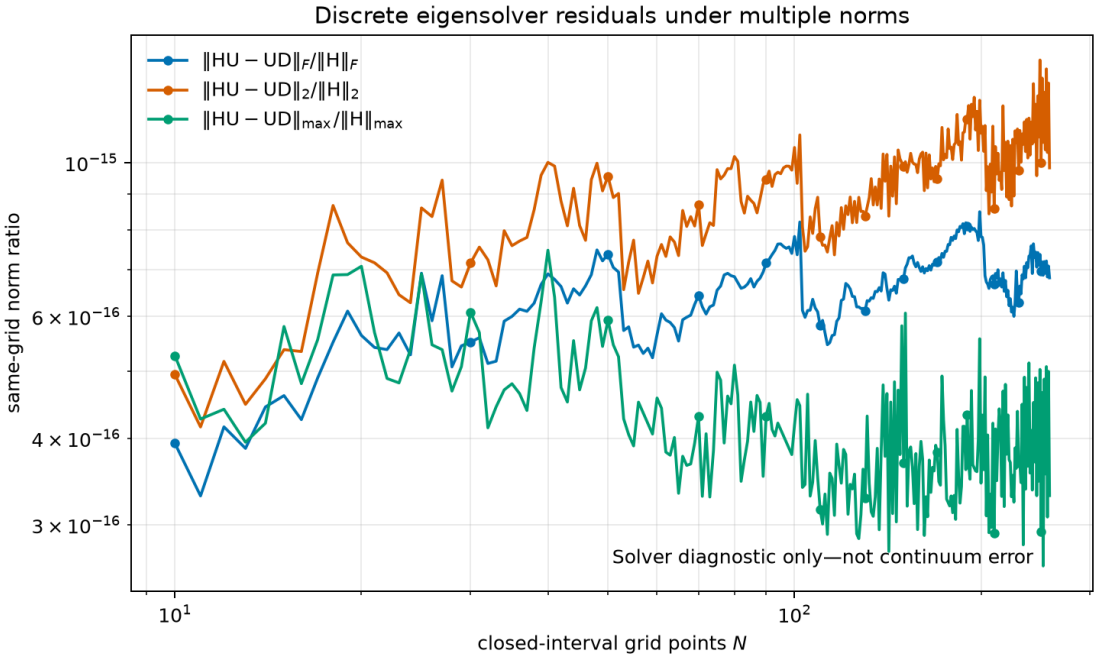


Figure 8: Algebraic eigensolver residual ratios under three matrix norms. These are not continuum-error estimates.

8 Residual identifiability

The reduced Hamiltonian does not uniquely identify a kinetic–potential partition. The ambiguity is structural rather than numerical. If one admissible decomposition is $\mathbf{H}_{\mathrm{red}} = \mathbf{T}_0 + \mathbf{R}_{T_0}$, then every Hermitian matrix $\mathbf{K}$ on the retained space generates another exact decomposition,

$$\mathbf{H}_{\mathrm{red}} = (\mathbf{T}_0 - \mathbf{K}) + (\mathbf{R}_{T_0} + \mathbf{K}). \tag{28}$$

The set of decompositions is therefore an affine family unless prior physical conditions restrict either term. On the $M = 3$ retained coordinate space, the declared physical box decomposition is

$$\mathbf{H}_{\mathrm{red}} = \mathbf{T}_{\mathrm{D},red} + \mathbf{0}. \tag{29}$$

For a declared illustrative real-symmetric shift $\mathbf{K}$, the same matrix also satisfies

$$\mathbf{H}_{\mathrm{red}} = (\mathbf{T}_{\mathrm{D},red} - \mathbf{K}) + \mathbf{K}. \tag{30}$$

The physical decomposition reconstructs with zero Frobenius error. The shifted decomposition reconstructs with error 1.11×10^{-16}. This agreement is expected from the algebra and verifies only its finite implementation. The second equality does not make $\mathbf{K}$ a physical potential. A physical assignment would require independent justification of the shifted kinetic reference and of the admissible structure of the remaining term.

9 Dependence on the admissible model class

The illustrative shift was fitted in Frobenius norm to four nested model classes. If $\mathcal{M}_V$ is the chosen class, the unexplained residual is

$$\mathbf{E}_{\mathcal{M}} = \mathbf{K} - \mathbf{V}^{\star}_{\mathcal{M}}. \tag{31}$$

Under the Frobenius inner product, each fit is an orthogonal projection onto the corresponding linear matrix subspace. The scalar fit retains only the mean trace, the diagonal fit retains all diagonal entries in the declared basis, and the tridiagonal fit additionally retains nearest-coordinate couplings. The unrestricted real-symmetric class contains the target itself. This hierarchy makes two points explicit: model-class restriction creates the informative residual, and basis-dependent classes such as "diagonal" or "tridiagonal" are meaningful only after the retained basis has been identified.

Table 7: Frobenius residuals after fitting the illustrative shift.

Admissible model class	$\|\mathbf{E}_{\mathcal{M}}\|_F$
Scalar identity	1.9967
Diagonal in retained basis	1.8561
Real symmetric tridiagonal	0.5657
Arbitrary real symmetric	0.0000

The unrestricted class fits exactly by construction and is therefore uninformative. Restricted classes leave different unexplained residuals. The scientific inverse problem begins only after locality, range, symmetry, differential form, parameter sharing, transferability, or another admissible structure has been specified.

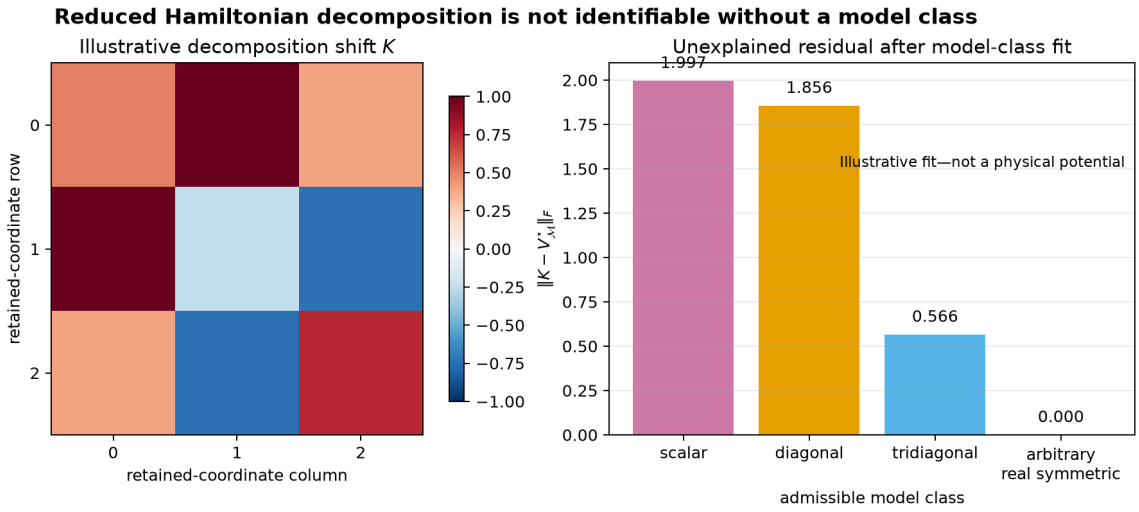


Figure 9: Illustrative decomposition shift and model-class dependence of the unexplained Frobenius residual. The shift is not assigned as a physical potential.

10 Discussion

The calculations separate three error and interpretation questions that would otherwise be easy to conflate. First, the algebraic residual $\mathbf{H}_h\mathbf{V} - \mathbf{V}\mathbf{\Lambda}$ concerns solution of the finite eigenproblem. Its near-roundoff magnitude shows that diagonalization is accurate. Second, the difference $E_{n,h} - E_n$ concerns approximation of a specified continuum eigenvalue by the finite operator. Its second-order decay for fixed n is a discretization statement. Third, an operator residual such as $\mathbf{H}_{\mathrm{red}} - \mathbf{T}_0$ concerns a chosen decomposition. Its physical meaning is not determined by either of the first two error measures.

The norm study sharpens this separation. The boundary residual is sparse and therefore small relative to the aggregate Frobenius size of the growing Hamiltonian, yet its spectral action remains one quarter of the Hamiltonian scale asymptotically. Neither statement is more correct; each answers a different question. A model-reduction criterion must therefore declare whether it controls average matrix content, worst-case action, selected states, or another operational quantity. Reporting an unlabeled norm would suppress this choice.

The unmatched-compression calculation provides a complementary warning. Its order-one normalized norm does not show that spectral retention has failed. It shows that the retained and reference terms were subjected to different maps. For observables restricted to the retained eigenspace, the discarded block may be intentionally irrelevant; as a full-space operator subtraction, however, it is necessarily large. The state space on which a claim is made is therefore part of the claim itself.

Finally, the model-class experiment converts the nonuniqueness proof into a finite calculation. Exact reconstruction by multiple decompositions is not a competition between physical models: one alternative was introduced solely to exhibit the affine ambiguity. Scientific content enters only when external principles restrict the kinetic reference and potential class. In later semiconductor reductions those restrictions may involve locality, symmetry, range, shared parameters, or transferability, but this illustrative example does not select among them.

11 What the combined study establishes

The calculated illustrative experiment establishes the following finite statements under the retained inputs and binary64 implementation:

1. The centered finite-difference spectrum agrees with its closed-form discrete oracle.
2. Fixed-index eigenvalues converge at the expected second order once resolved.
3. The growing full spectrum is not uniformly accurate.
4. Consistent reduction gives the zero physical potential for the box.
5. Unmatched reduction gives the negative discarded kinetic sector.
6. The declared boundary subtraction gives a finite boundary-reference operator.
7. Frobenius, spectral, and maximum-entry norms emphasize different aspects of the same residual.

8. Accurate solution of the discrete eigenproblem does not imply continuum accuracy.
9. The same reduced Hamiltonian permits distinct exact algebraic decompositions.
10. The unexplained residual depends on the admissible model class.

The central conclusion is not that one residual norm is preferable in every circumstance. A residual can be interpreted only after the state spaces, maps, reference operator, norm, and admissible model class have been declared.

12 Limitations

- The cyclic closure is a declared finite comparison reference, not a unique continuum reference operator.
- Same-grid normalization does not align operators across changing numerical spaces.
- Fixed-mode convergence does not establish uniform spectral or operator convergence.
- Nodal eigenvector overlap does not establish continuum interpolation error.
- The illustrative shift $\mathbf{K}$ has no physical assignment.
- The model-class comparison uses a declared retained basis and Frobenius metric; changing either changes the fit.
- No result validates a material model or supplies evidence about silicon, DFT, Wannier localization, or impurity physics.

13 Reproduction and retained evidence

The exact commands are recorded in `README.md`. The retained evidence consists of frozen JSON inputs, standalone calculation scripts, independent verification scripts, retained JSON results, calculated graphics, software identities, and SHA-256 checksums. Passing verification establishes agreement with the stated finite mathematics and analytical oracles. It does not provide scientific validation or human acceptance of a physical model.

References

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